Measurement of Gravitational Constant With Geometric Corrections of a Cavendish Torsion Balance

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A torsional Cavendish Balance was used to measured the gravitational constant G using three distinct methods. For each method, G values are reported assuming both an ideal torsion balance geometry and an asymmetric geometry based on observations of the physical apparatus used to take data. The most precise method fits a damped sinusoidal curve to an oscillating rod using the corrected geometry, finding $G = 6.50 \pm 0.39 \times 10^{-11}$ m^3 kg^{-1} s^{-2}. Other historical and modern measurements of G are discussed and used to analyze the limitations of the methods used in this paper.

I. INTRODUCTION

The gravitational force F_G is a fundamental physics concept that is taught early on to high school physics students as an introduction to classical mechanics. Despite how simple the gravitational force may seem to be, measuring G, the gravitational constant that governs this force, has historically been a challenge due to how weak this force is compared to others (such as the electrostatic force) and how sensitive this force is to small changes in geometry.

G was first measured by Cavendish in 1798 using a torsion balance, suspending two smaller masses from a thin copper wire that would rotate when larger masses were placed nearby. Cavendish was able to adjust the position of the larger masses and measure the deflection of the smaller masses using vernier scales; this was then used to determine the force being applied to the rod and thus calculate G [1]. After his original measurement, other physicists continued to iterate on the design of his experiment, including Boys (who used a more sensitive quartz fiber and smaller apparatus), and Poynting (who replaced the torsion balance with a common beam balance) [2, 3]. Although these early experiments could only measure G to within roughly 0.5% accuracy, modern physicists have continued to design other apparatuses that have reduced this uncertainty by multiple orders of magnitude [4].

This experiment aims to utilize a traditional torsional Cavendish Balance to measure G, but with the support of more modern technology. Specifically, rather than relying on vernier scales, a laser-and-mirror setup can help magnify the deflection of the beam by acting as a large lever arm. The projection of this laser can then be measured digitally using a video recorder, providing a finer precision than just simply recording deflections by hand using a meter stick.

As with previously-conducted measurements of G, applying all of the possible corrections for the apparatus used is difficult, but determining which corrections are most crucial can help greatly mitigate overall uncertainty. For example, in 1874, Crookes analyzed how temperature gradients between the masses could create small attracting and repelling forces, which would become magnified if large enough masses were used [5]. Poynting also goes into depth about the all of the corrections Cavendish either applied in his experiment or deemed small enough to ignore, including the attracting masses being offset, the attraction to the rotating rod itself, and even the attraction to the surrounding frame and scales [3]. Most of these corrections individually adjusted position or frequency measurements by less than 1%, but considering all of these corrections simultaneously is what allowed these early physicists to reduce their overall error. Measuring G with different types of apparatuses or forms of measurement also helped ensure that other systematic errors were properly accounted for.

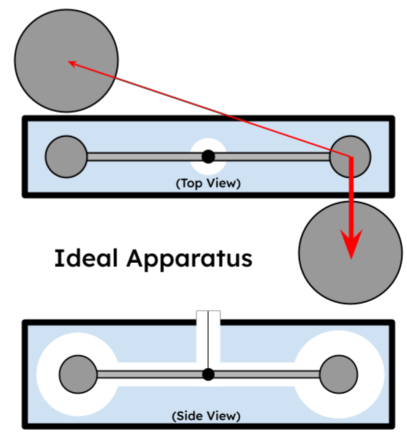

FIG. 1. A diagram of an ideal torsional Cavendish Balance. Both a top view (above) and a side view (below) are provided. Red arrows represent the gravitational force vectors felt by one of the smaller masses. For this experiment, an ideal Cavendish Balance is perfectly symmetric, with a central rod that runs parallel to the walls of the encapsulating box (shown in blue) and is centered in the middle of the box. The two larger masses are attached to a rotating plate with a rotation axis centered on the rod's pivot point.

A similar experimental approach will be taken with the data and corrections taken in this paper. The same apparatus will be used with different measurement methods to determine G, and the apparatus will be analyzed for sources of uncertainty so that the largest sources can be corrected and accounted for.

The total gravitational force (perpendicular to the rod) felt by one of the smaller masses in the ideal Cavendish Balance setup can be calculated as

$$F_{\perp} = \frac{GmM}{b} - \frac{GmM \cdot b}{(4d^2 + b^2)^{3/2}} \tag{1}$$

where G is the gravitational constant, m is the mass of one of the smaller ball attached to the rod, M is the mass of one of the larger balls, b is the distance from a small mass to the nearest large mass, and d is the distance from the rotation axis to a small mass. Since acceleration is measured in this experiment, the small mass m does not need to be known. In addition, factoring G and M leaves a term just consisting of the geometrical terms b and d. When applying small corrections to the apparatus, this geometrical term is the only place where the calculation will be modified.

Since the small masses only move a very small amount across the duration of the measurements (less than 0.1 mm), we can make use of small-angle approximations to relate the period T of oscillations to the inertia I of the rod and the torsion constant κ, via

$$T^2 = \frac{4\pi^2 I}{\kappa}. \tag{2}$$

Since the torque of the wire $\tau = \kappa\theta$ where θ is the angle of the rod, and the torque must also equal $F_{\perp}d$, we can use the period and displacement measurements of the rod to calculate G.

II. APPARATUS AND DATA COLLECTION

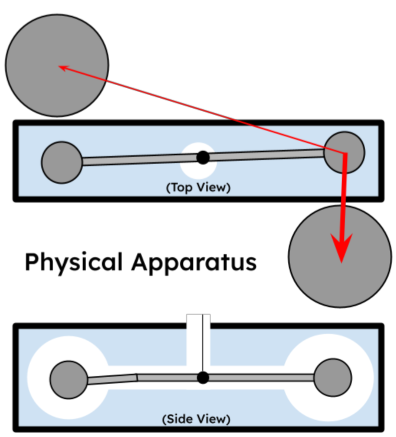

FIG. 2. A diagram of the physical (real) Cavendish Balance used for this experiment. There are three major asymmetries in the physical apparatus not present in the ideal balance: 1) The central rod is not perfectly parallel to the encapsulating box, but is rotated counter-clockwise slightly 2) The rod's pivot point is offset from the rotation axis of the larger masses 3) One side of the rod is bent downwards. The first two asymmetries are corrected for in the curve fitting measurements of G.

Fig. 2 shows a diagram of the physical Cavendish Balance that was used for this experiment, including its central axis offset, starting rotation, and bent rod. This central axis is shifted by roughly 3.2 mm, and the starting rotation is only about 3° off from the horizontal. A mirror attached to the rod redirects light from a laser into a wall with a thin meter stick. A camera (nearly aligned with the incoming laser beam) is placed around 3 meters from the wall, with the viewing angle as perpendicular as possible. The projected "dot" of the laser beam was then tracked to determine the rotation of the rod.

All videos and photos taken of the laser dot were processed using the Tracker software. Frames were preprocessed using the "Perspective Filter" to mitigate any skewed-angle effects of the camera. Video footage also used the "Baseline Filter" to subtract a background image with no laser dot to make tracking easier. Finally, the meter stick attached to the wall was used to establish a pixel-to-meter scale. The "Autotracker" feature was then used to determine the position of the laser dot in any given frame using an initial key frame. Videos were taken at 30 fps, and every 5 frames were analyzed for a time resolution of 0.17 s.

The gravitational constant G was measured using 3 distinct methods: curve fitting an initial acceleration, curve fitting a damped sinusoidal, and final displacement. In both curve fitting methods, the large masses were swung from one side of the box to the other, generating the largest change in force possible on both masses, and video of the motion was recorded. In the final displacement method, the large masses were rotated to different angles (spaced 2.5° apart) until the rod came to rest, and photos of the rotation were recorded. For all methods, G was separately measured with no geometric corrections (assuming the ideal apparatus in Fig. 1) and with some of the most dominating observed corrections due to the asymmetry (as seen in Fig. 2). More specifically, the angle and rotation axis offsets were factored into the calculations, but the bending of the rod was ignored (as well as any other effects, such as thermal).

III. RESULTS

Fig. 3 shows a plot of the position of the laser dot over the first 50 seconds after the large masses had been moved. A quadratic fit was then applied to the data to determine α, the initial angular acceleration of the rod. Attempting to make the time window larger than 50 seconds for the initial acceleration measurement showed the data diverging from the quadratic trend, so this was chosen as the largest window that still falls within a clearly quadratic regime. Although there is clear indications of noise on the millimeter scale between individual measurements, the overall quadratic trend appears to describe the data well within this timescale. Data was collected for the cases where the large masses were rotated clockwise and counterclockwise, then G was calculated separately for each measured α and then averaged between the two cases. This was repeated for both the ideal/symmetric geometry as well as the physical/asymmetric geometry using the corrections described in Sec. II.

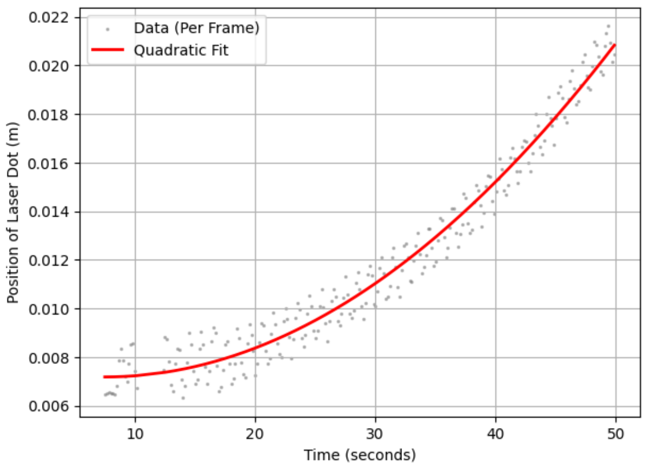

FIG. 3. A plot of the position of the laser dot after the large masses were rotated as a function of time. A short time scale of 50 seconds was used, and a quadratic curve was fit to the data. Six measurements were taken each second, so each data point is separated by 0.2 seconds. The quadratic curve was used to extract an angular acceleration α of the rod.

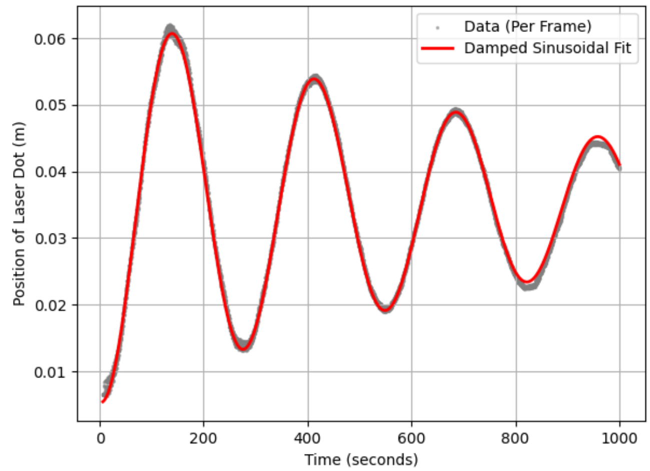

FIG. 4. A plot of the position of the laser dot, fit to a damped sinusoidal curve. Unlike with the quadratic fit, data was taken over the course of 1000 seconds with the same framerate. After the first 40 seconds, the noise in the measurements appears to decrease. The fit damped sinusoidal curve was used to extract an angular frequency ω and a damping constant γ.

Fig. 4 is similar to the previous quadratic fit plot, except it fits a damped sinusoidal to a larger time window of 1000 seconds. At very early times (first 40 seconds), it appears the noise is somewhat large, but quickly diminishes to produce a much tighter band of data. The damped sinusoidal fit the to the curve tightly follows the data, especially when the dot is moving at a high speed. The peaks in the data do not appear to fit quite as closely to the model, but overall the model appears to be a good predictor of the data. An angular frequency ω and a damping constant γ were extracted from this curve and used to calculate G. Again, data was taken for the apparatus in two configurations, and G was calculated independently for both configurations and then averaged together assuming both ideal and physical geometries.

Lastly, the angles measured for the final displacement method were converted into distances b between the large and small masses (assuming ideal geometry) and a linear regression was performed to determine G. The R^2 value for this regression was then converted to a percent error (35%) and applied to the measured G value. All of the measured G values for this experiment, as well as some additional reference values, are reported below in Table III.

The curve-fitting methods used for this experiment appear to be significantly more reliable than the final displacement method, likely due to the noise in position measurements for a single frame. From the video footage, it is clear that the mirror was prone to some fast oscillations which would shake the laser dot a millimeter or two off of a central value. Although most measurements were taken during non-primary lab times (to mitigate the number of people moving around the experiment), just walking around in the same room as the Cavendish Balance was enough to cause this minor shaking, which was unavoidable when collecting data. This was especially prevalent when adjusting the angle of the large masses, since spinning them right below the hung masses would instantly generate a lot of shaking, sometimes enough to halt data collection for a few hours. This is likely why Cavendish used pulleys and telescopes in his original experiment so that he could view the readings of his vernier scales from afar [1]. Video footage had enough frames taken over a short time scale to average-out this noise to a clear central value, but the single photos used for the final displacement were simply too noisy to measure a value for G beyond an order of magnitude. This is why only the ideal calculation was done for the final displacement method, since the errors due to the fitting were much larger than the geometric corrections done for the apparatus.

Since the initial acceleration (quadratic fit) curve is on a shorter timescale and is only an approximation of the damped sinusoidal model, the initial acceleration measurements can not be any more accurate than the damped oscillator. This is also evidenced by the relative noise between the two measurements; the quadratic fit in Fig. 3 is centered on a much thicker band of noise than the fit in Fig 4. However, at least for the symmetric geometry, the uncertainty in the position of the small masses within the apparatus completely dominate the uncertainty of the fits between these two curve fitting methods.

The geometric corrections of the apparatus had a mas-

Method	Measured G (10^{-11} m^3 kg^{-1} s^{-2})
Initial Accel. (Symm)	7.03 ± 1.9
Initial Accel. (Asymm)	6.82 ± 0.43
Damped Osc. (Symm)	5.71 ± 1.4
Damped Osc. (Asymm)	6.50 ± 0.39
Final Displacement (Symm)	2.45 ± 0.86
CODATA Reference	6.67430 ± 0.00015 [4]
Cavendish (1798, corrected)	6.74 ± 0.04 [3]

TABLE I. A collection of all the G values measured using the initial acceleration, damped oscillator, and final displacement methods alongside reference values. For both curve fitting methods, two values of G are presented: one using symmetric Cavendish Balance geometry (shown in Fig 1), and another using asymmetric geometry used to correct for the physical apparatus used (shown in Fig 2. For Cavendish's reference value, his measurement was converted from being a factor of the density of water to the standard units of G, and the corrected value of his data was used rather than the one found in his original publication due to a transcription error [3].

sive impact on the reported G values, so a lot of care was taken to ensure that the corrections chosen were consistent with photographs of the apparatus and not biased by other referenced G values. Initially, it was assumed that the rod was well-centered and just had some initial rotation that was not parallel to the sides of the box. This observation was what encouraged taking measurements for both clockwise and counter-clockwise rotation of the large masses, since different α, ω, and γ should be obtained given this asymmetry. After closer inspection of the apparatus with others, the off-axis nature of the rod was also discovered, so the calculations were updated to reflect this shift. Between the symmetric, first correction, and final correction measurements for G, the measured values increasingly converged, an indication that these corrections were more accurately describing the apparatus. The offset values that were measured were set before they were plugged into the G calculations so that they would not just be tweaked to match a particular reference value.

Other undergraduate torsion Cavendish Balance experiments have aimed to increase the sensitivity in measuring the rotation of the rod by using a line of phototransistors, which are more sensitive to changes in the intensity of the laser dot than a video or photo would be [6]. Although more accurate in measuring the position of the dot, other systematic uncertainties still dominate their error, which was reported around 5%. Given this, it appears the key limitation of a torsion balance is still in the geometry of the masses and rods themselves, and not the accuracy of measuring the rod's deflection.

As seen in Table III, the two G values that used the physical geometry corrections are in agreement with the reference values and with much smaller uncertainty than when the ideal geometry was used. The initial acceleration method appeared to initially overshoot G, and the geometrical corrections reduced it some, whereas the damped oscillator method undershot G and the corrections increased it. In the symmetric cases, the uncertainty in b vastly dominated any other sources, but once corrected for, was reduced to a similar scale of the uncertainty sourced from the radius R of the large masses. Since these large masses were not particularly spherical and contained a lot of dents from use over time, their circumferences were measured many times at different angles and then averaged. Since this radius R was then cubed to determine the mass M of the large masses, it is another sensitive quantity for measuring G.

From more modern experiments that have measured G, it appears that the geometrical limitations described above are especially difficult to surmount, which is why newer apparatuses have used setups with distances of much higher tolerances. For example, Gundlach and Merkowitz did not used the standard dumbbell design, but rather carefully machined plates which held their attracting masses in place [7]. They also used a torsion fiber, but made use of a motor to adjust the rotation of the masses so that the actual deflection of the fiber was always near zero. The apparatus was also under heavy temperature control to avoid thermal expansion. Most recently, in 2018, similar methods were used with even more advanced torsion fibers (silica with metallic coatings used to suppress non-elastic behaviors of the fiber) to reduce uncertainty below 1 part in 10 million [8]. Regardless, these other corrections only become relevant when the positions of all massive objects are known very precisely, which is simply not possible for the torsion balanced used to collect the data in Table III.

IV. CONCLUSION

The geometric corrections used in this experiment are critical for reducing the overall uncertainty of the measured G values. From careful observation of the apparatus alone, the relative error was reduced from ∼20% to ∼6%, and the corrected values were in agreement with other reference values. The damped sinusoidal curve fitting method ($G = 6.50 \pm 0.39 \times 10^{-11}$ m^3 kg^{-1} s^{-2}) was shown to be the most accurate measurement conducted of the three due to the bulk of video footage frames, capable of seeing through the noise of fast oscillations due to vibrations in the building around the apparatus. These uncertainties were reduced to an extent where a more advanced torsion balance would be required to increase accuracy any further.

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